Hydrogen Ion Concentration and pH

From Kw to the pH scale — measuring acidity precisely

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Standards

Chemistry 2.6 — Acid–Base Chemistry: Kw, pH, and pOH calculations.

Objectives

1. What is the ion product constant for water, Kw?
2. How are pH and pOH defined and related?
3. How do we calculate $[H^+]$ or $[OH^-]$ from pH?

Expectations

1. Calculate pH, pOH, $[H^+]$, and $[OH^-]$ from one another
2. Place common substances on the pH scale

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What Do You Already Know?

WHAT I KNOW	WHAT I WANT TO KNOW	EXPLORE THE SOURCE
▪ Lemon juice and battery acid are both “acidic” ▪ Soap and ammonia are both “basic” ▪ Blood stays right around pH 7.4	▪ How is pH actually calculated? ▪ Why is the pH scale logarithmic? ▪ What does a “negative log” even mean?	► PhET: Acid-Base Solutions ► PhET: pH Scale ► Khan Academy: Acids & Bases *Tap a link to explore before class discussion.*

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The Ion Product Constant for Water, Kw

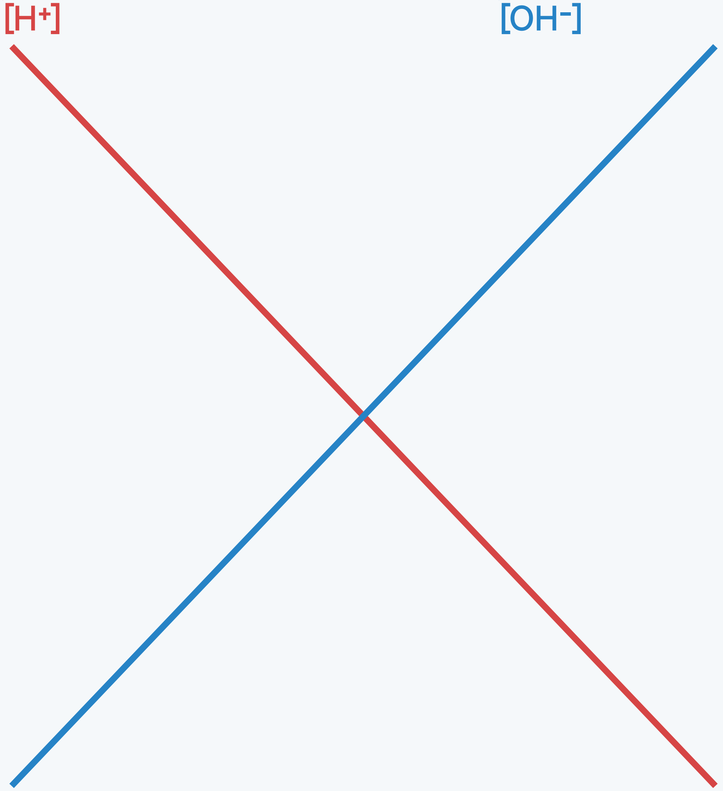
- Pure water contains equal $[H^+]$ and $[OH^-]$.
- $K_w = [H^+][OH^-]$ — at 298 K, $K_w = 1.0 \times 10^{-14}$
- Le Châtelier's Principle: as $[H^+]$ rises, $[OH^-]$ must fall so the product always equals Kw.

Worked Example

A cup of coffee has $[H^+] = 1.0 \times 10^{-5}$ M. Find $[OH^-]$.

$[OH^-] = K_w / [H^+] = (1.0\times10^{-14})/(1.0\times10^{-5})$

$[OH^-] = 1.0 \times 10^{-9}$ M → coffee is acidic



As one ion rises, the other falls — their product stays fixed at Kw.

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pH and pOH

$pH = -\log[H^+]$

The negative logarithm of the hydrogen-ion concentration.

$pOH = -\log[OH^-]$

The negative logarithm of the hydroxide-ion concentration.

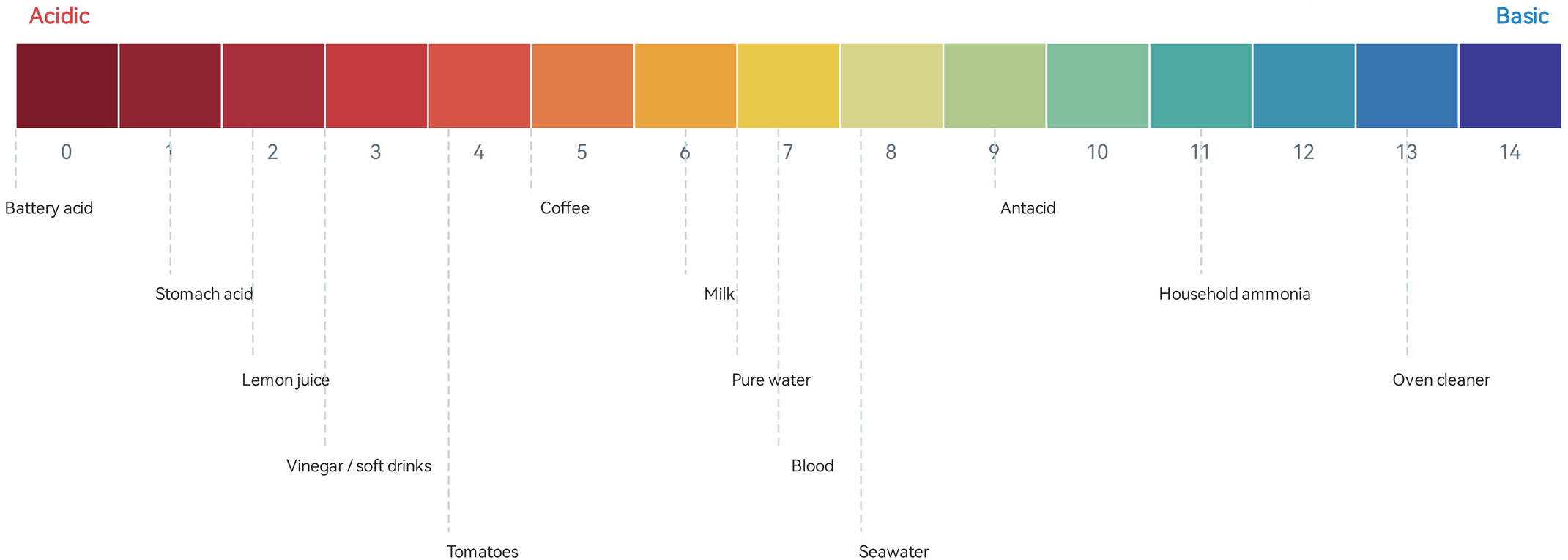
$pH + pOH = 14.00$

At 298 K, the two always sum to 14.

- As $[H^+]$ increases, pH decreases (lower pH = more acidic).
- pH < 7 acidic • pH = 7 neutral • pH > 7 basic (at 298 K)
- Because pH is a log scale, each whole step is a 10× change in $[H^+]$.

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The pH Scale in Everyday Life



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Worked Example: pH of Blood

What are $[H^+]$ and $[OH^-]$ in blood at pH 7.40 (298 K)?

Find $[H^+]$

$pH = -\log[H^+]$

$-pH = \log[H^+]$

$[H^+] = \text{antilog}(-pH)$

$[H^+] = \text{antilog}(-7.40)$

$$[H^+] = 4.0 \times 10^{-8}\ M$$

Find $[OH^-]$

$pOH = 14.00 - 7.40 = 6.60$

$pOH = -\log[OH^-]$

$[OH^-] = \text{antilog}(-6.60)$

$$[OH^-] = 2.5 \times 10^{-7}\ M$$

Try it yourself: formic acid at 0.100M has pH = 2.38 — what is Ka for HCOOH?

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Lesson Summary

Key Takeaways

1. Kw = $[H^+][OH^-] = 1.0 \times 10^{-14}$ at 298 K — as one ion concentration rises, the other falls.

2. pH = $-\log[H^+]$ and pOH = $-\log[OH^-]$; pH + pOH = 14.00.

3. The pH scale is logarithmic — each unit represents a 10-fold change in $[H^+]$.

4. Given any one of pH, pOH, $[H^+]$, or $[OH^-]$, you can calculate the other three.

Test Yourself

1. Calculate the pH of solutions with $[H^+] = 1.0 \times 10^{-2}$M and $[H^+] = 3.0 \times 10^{-6}$M at 298 K.

2. Calculate the pH and pOH of a solution with $[OH^-] = 6.5 \times 10^{-4}$M.

3. Calculate $[H^+]$ and $[OH^-]$ for milk (pH = 6.50) and household ammonia (pH = 11.90).

4. Challenge: calculate $[H^+]$ and $[OH^-]$ in a sample of seawater with pOH = 5.60.

1- $pH = -\log[H^+]$

$= -\log[1 \times 10^{-2}] = 2$

2- $pH =$

$pOH = -\log[6.5 \times 10^{-4}] = 3.1$

$pH + pOH$

$pH + 3.1 = 14$

$pH = 14 - 3.1 = 10.9$

Basic

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acid ← → base

0 — 7 — 14

$pH = -\log [H^+]$

$pOH = -\log [OH^-]$

$pH + pOH = 14$

$[H^+] = \text{antilog} - pH$

$[OH^-] = \text{antilog} - pOH$

$Kw = [H^+][OH^-] = 1 \times 10^{-14}$